

Caifeng Zou

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Seismological Laboratory
California Institute of Technology
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EDUCATION

California Institute of Technology 2022 –

Ph.D. Geophysics (Minor in Computational Science and Engineering)

Advisors: Zachary E. Ross & Robert W. Clayton

Imperial College London 2021 – 2022

M.Sc. Applied Computational Science and Engineering

Grade: Distinction

Tongji University 2016 – 2020

B.Sc. Geophysics

GPA: 94.32%

PUBLICATIONS

1. **Zou, C.**, Shi, Y., Ross, Z. E., Clayton, R. W., & Azizzadenesheli, K. (2026). Enforcing Reciprocity in Operator Learning for Seismic Wave Propagation. *Seismological Research Letters*.
2. Kong, Q., **Zou, C.**, Choi, Y., Matzel, E. M., Azizzadenesheli, K., Ross, Z. E., Rodgers, A. J., & Clayton, R. W. (2026). Reducing Frequency Bias of Fourier Neural Operators in 3D Seismic Wavefield Simulations through Multistage Training. *Seismological Research Letters*, 97(1), 272-282.
3. **Zou, C.**, Ross, Z. E., Clayton, R. W., Lin, F. C., & Azizzadenesheli, K. (2025). Ambient Noise Full Waveform Inversion with Neural Operators. *Journal of Geophysical Research: Solid Earth*, 130(11), e2025JB031624.
4. **Zou, C.**, & Clayton, R. W. (2025). Imaging the Northern Los Angeles Basins with Autocorrelations. *Seismological Research Letters*, 96(3), 1791-1801.
5. Zheng, H., Chu, W., Zhang, B., Wu, Z., Wang, A., Feng, B., **Zou, C.**, Sun, Y., Kovachki, N. B., Ross, Z. E., Bouman, K., & Yue, Y. (2025). Inversebench: Benchmarking Plug-and-Play Diffusion Models for Scientific Inverse Problems. *In The Thirteenth International Conference on Learning Representations*.
6. **Zou, C.**, Azizzadenesheli, K., Ross, Z. E., & Clayton, R. W. (2024). Deep Neural Helmholtz Operators for 3D Elastic Wave Propagation and Inversion. *Geophysical Journal International*, 239(3), 1469-1484.
7. **Zou, C.**, Zhao, L., Hong, F., Wang, Y., Chen, Y., & Geng, J. (2023). A Comparison of Machine Learning Methods to Predict Porosity in Carbonate Reservoirs from Seismic-Derived Elastic Properties. *Geophysics*, 88(2), B101-B120.
8. **Zou, C.**, Zhao, L., Xu, M., Chen, Y., & Geng, J. (2021). Porosity Prediction with Uncertainty Quantification from Multiple Seismic Attributes Using Random Forest. *Journal of Geophysical Research: Solid Earth*, 126(7), e2021JB021826.

9. Zhao, L., **Zou, C.**, Chen, Y., Shen, W., Wang, Y., Chen, H., & Geng, J. (2021). Fluid and Lithofacies Prediction Based on Integration of Well-Log Data and Seismic Inversion: A Machine-Learning Approach. *Geophysics*, 86(4), M151-M165.

INVITED TALKS

Seismic Wavefield Modeling and Inversion Using Neural Operators SIAM Conference on Imaging Science	Nov, 2026
Ambient Noise Full Waveform Inversion with Neural Operators UCLA Planetary/Geophysics/Geochemistry Seminar	Feb 5, 2026

PROFESSIONAL SERVICE

Reviewer for <i>Geophysical Journal International</i> (3 reviews) <i>Geophysics</i> (10 reviews) <i>Geophysical Prospecting</i> (2 reviews) <i>Earth and Space Science</i> (2 reviews) <i>Journal of Geophysical Research: Machine Learning and Computation</i> (1 review)	
Organizer for the monthly Caltech Seismo Lab Coffee Hour on Machine Learning	2026
Co-convenor for session Developing Data-Driven Methods in the AI Era: New Approaches to Earthquake Science Seismological Society of America Annual Meeting (Pasadena, CA)	2026
Co-convenor for session Advances in Machine Learning for Solid Earth Geoscience American Geophysical Union Annual Meeting (New Orleans, LA)	2025
Organizer for the weekly Caltech Seismo Lab Seminar	2024 – 2025

TEACHING

Machine Learning in Geophysics	Teaching Assistant	Caltech	FA 2025 – 26 FA 2024 – 25
Projects in Machine Learning	Teaching Assistant	Caltech	FA 2024 – 25
Seismology	Teaching Assistant	Caltech	SP 2023 – 24